Improving Clinical Information Extraction with Public-Data Pretraining

Extended Summary

Doctoral thesis · Sorbonne Université

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Contents

Part I

Introduction

> *"The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin."*
>
> Richard S. Sutton, *The Bitter Lesson*

How can we teach an artificial intelligence to read hospital notes when hospital notes cannot be used to teach it? The notes themselves would be the best examples because they contain the words, shortcuts, and clinical habits the system needs to learn. But they describe real patients, so they stay inside the hospital, and a system trained on them may have to stay there too. This thesis asks whether public biomedical text can be used instead.

The question is old. In 1959, Robert Ledley and Lee Lusted set a medical goal for the new field of artificial intelligence: a computer that could weigh a patient's signs toward a likely disease (Ledley and Lusted, 1959). Their question is also the question of this thesis: where does the knowledge that drives such a system come from, and can a machine be given enough of it to be useful in care? The early medical systems answered from their own records. De Dombal's acute-abdomen system reached 91.8% diagnostic accuracy on 304 patients at a single centre in 1972 (de Dombal et al., 1972), but when it was taken to a multicentre trial across eight centres and 16,737 patients, accuracy with the aid in place rose only to 65.3% (Adams et al., 1986). A system tuned on Leeds patients did not carry its 91.8% to other hospitals, because the probabilities it relied on came from one place and the new patients came from many. A learning system depends on the data its knowledge was built from, so a result measured in one setting may not hold in another.

In the 1970s, medical AI shifted toward *expert systems*, which stored specialist knowledge as explicit rules. MYCIN, at Stanford, advised on antibiotics through IF–THEN rules (Shortliffe and Buchanan, 1975); in a blind test, eight infectious-disease specialists judged it acceptable in 65% of cases, against a faculty mean of 55.5% (Yu et al., 1979). It passed the benchmark and was never used to treat a patient (Miller et al., 1982). A program could win a blind benchmark and still be unusable, because winning the benchmark was not what physicians were asking for.

Machine learning later changed how systems acquired knowledge. Instead of being told the rules, systems learned what to do from large sets of examples, and the gains were large: by 2016 a network trained on 128,175 retinal photographs read them for diabetic eye disease about as accurately as ophthalmologists (Gulshan et al., 2016). In the biomedical literature, expert-system papers peak around 1991, then a deep-learning line near zero before 2015 climbs past twenty thousand papers a year. These results raised a new requirement, the one that organises this thesis: a system that learns from examples needs many well-matched examples, and clinical text does not circulate freely.

Most modern natural language processing starts by pretraining a language model on large collections of raw text, before adapting it to a task. In the clinical setting this path is hard to follow. Real clinical notes are personal data and cannot be shared, so France has no public corpus of them to pretrain on. A model trained inside one hospital on its own records carries the same constraint and cannot be released to others. A shareable clinical model has to be built from public text.

This thesis asks one question: can public data support language models for French clinical information extraction, without training on sensitive clinical records? The manuscript answers it through three questions.

- Can we assemble a useful public French biomedical corpus, and does its content and quality change what the model learns?
- Given such a corpus, how should we pretrain a language model on it: by adapting an existing model, choosing the training objective, or adding knowledge from outside the text?
- How can we adapt these models to clinical extraction tasks when labelled clinical data is scarce, through model architecture and synthetic data?

This thesis makes seven contributions.

- *biomed-fr*, a public French biomedical corpus of 413 million words, used to train CamemBERT-bio. The adapted model finds biomedical terms in French text, such as drugs and diseases, more accurately than the general model it started from, with a 2.54-point average gain and at a fraction of the cost of training from scratch (Touchent and de la Clergerie, 2024).
- *Biomed-Enriched*, a labelling of the open biomedical literature (PubMed Central) by content type and quality. It identifies about two million paragraphs of clinical cases and improves a model trained further on them (Touchent et al., 2026).
- *MC-Bio*, a ten-billion-token French biomedical corpus, and a study of which content types and quality signals help. The experiments show that some common filters degrade performance.
- A pretraining method that trains the model on a different task for a while and then switches back to the main objective, improving the model at no extra cost.
- *OntoBook*, a way to add structured medical knowledge to pretraining by turning a medical knowledge base into training text.

- Model designs, frenchmed-gliner and MedEmbed, that find clinical terms from only a handful of labelled examples.
- A method for generating synthetic clinical reports by rewriting public biomedical text into hospital style, so that extractors can be trained without access to real hospital data.

The parts follow these questions. The first asks what biomedical text is and how far its public forms are from the clinical notes we care about. The second builds and studies a public French biomedical corpus. The third pretrains language models on it, through model adaptation, training objectives, and knowledge enrichment. The final part adapts these models to clinical extraction tasks under scarce labelled data. The conclusion then opens onto a question this work leaves behind: once such models are built, how they behave when someone actually uses them becomes its own problem.

Part II

What Is Biomedical Text?

FRANÇAIS

CENTRE HOSPITALIER DE SAINT-AUBIN
Dossier n° 4471902
Cardiologie, Soins Intensifs
Séjour : 03/11–08/11/2024

Patient : CARON Julien, né le 14/03/1962 (62 ans). **Dest. :** Dr S. Lemaire.

Motif. Douleur thoracique constrictive rétrosternale, irradiation bras G, depuis 2h.

Antécédents. HTA. DT2 sous metformine. Tabac 30 PA. Dyslipidémie.

Examen. TA 148/92, FC 98, SpO_2 96% AA, EVA 7/10. Pas de signe d'IVG.

Paraclinique. ECG : sus-décalage ST V1–V4. Tropo T 4520 ng/L (N <14). Coro : occlusion IVA proximale, angioplastie + stent actif, TIMI 3. ETT : FEVG 40%.

Au total. SCA ST+ antérieur (I21.0).

Ttt sortie. Kardégic 75, ticagrélor 90x2, atorvastatine 80, bisoprolol 2,5, ramipril 5.

CAT. Réadaptation cardiaque. Consultation cardio à 1 mois. Dr C. Després

ENGLISH · TRANSLATION

SAINT-AUBIN HOSPITAL
Record no. 4471902
Cardiology, Intensive Care
Stay: 03/11–08/11/2024

Patient: CARON Julien, b. 14/03/1962 (62 y). **To:** Dr S. Lemaire.

Reason. Constrictive retrosternal chest pain, radiating to L arm, for 2h.

History. HTN. T2DM on metformin. Smoking 30 PY. Dyslipidaemia.

Exam. BP 148/92, HR 98, SpO_2 96% RA, pain 7/10. No sign of LVF.

Workup. ECG: ST elevation V1–V4. Trop T 4520 ng/L (N <14). Angiography: proximal LAD occlusion, angioplasty + drug-eluting stent, TIMI 3. Echo: LVEF 40%.

Summary. Anterior STEMI (I21.0).

Discharge tx. Aspirin 75, ticagrelor 90x2, atorvastatin 80, bisoprolol 2.5, ramipril 5.

Plan. Cardiac rehabilitation. Cardiology review at 1 month. Dr C. Després

Figure 1: A discharge summary for an acute myocardial infarction; French original, English translation.

Figure 1 is a letter from one cardiologist to a colleague. For an ordinary French reader it is very hard to read: few verbs, telegraphic sentences, information dense and simply separated by commas like a table. This is not carelessness but a style of communication specially tuned for the community it is written for. The linguist Zellig Harris called such a specialised, rule-governed form a *sublanguage* (Harris, 1968), and the clinical note is its best-documented case (Sager, 1981). It is above all a form of compression: Anderson et al. (1975) showed that clinical shorthand obeys a small set of deletion rules, dropping only what the reader can put back. *Chest films normal* stands for *the chest films were normal*; what disappears is the part the expert reader already knows (Stalnaker, 2002). The note is also performative: a prescription lets the patient obtain the medication, an order to transfer a patient sets the transfer in motion (Austin, 1962).

All of this makes the note worth studying, but it also makes it hard to obtain. The note rarely leaves the hospital that produced it, because it is full of personal data, and removing the obvious identifiers does not fix this: a pseudonymised note is still not anonymous. For French the problem is worse. There is no shareable corpus of real French hospital records (Névéol et al., 2018). Public data is an alternative: published case reports, drug leaflets, article abstracts. But a case report is written to teach; it reads like a short story, in full sentences and the past tense. The same infarction, told in a research article or on a patient forum (Figure 2), is abundant and public, yet it speaks a different language.

FRANÇAIS	ENGLISH · TRANSLATION
Bonjour, mon père a fait une crise cardiaque la semaine dernière. Il a eu une grosse douleur dans la poitrine qui descendait dans le bras gauche, il était tout pâle et en sueur, on a appelé le 15 tout de suite. Est-ce que quelqu'un sait combien de temps dure la convalescence après un infarctus ? Merci d'avance.	*Hello, my father had a heart attack last week. He had a strong pain in the chest that went down into the left arm, he was all pale and sweating, we called emergency services right away. Does anyone know how long recovery takes after a heart attack? Thanks in advance.*

Figure 2: A patient forum about myocardial infarction; French original, English translation.

Biomedical text is not a single language. The same medical knowledge changes with who writes it, who reads it, and what the text is meant to do. Clinical notes, research articles, and patient-facing texts therefore offer different views of medicine, and only the latter two are widely available. The challenge for the rest of this thesis is to use this public text without losing sight of the clinical language we ultimately need.

Part III

Related Works

Natural language processing has converged, over recent years, on a single recipe: pretrain a model on large amounts of text, then adapt it to a task. BERT-style models (Devlin et al., 2019), built on the Transformer (Vaswani et al., 2017), established that recipe, and CamemBERT (Martin et al., 2020) carried it to French; the scaling of generative models (Brown et al., 2020; Touvron et al., 2023) is its continuation. All of it rests on one condition: text, in abundance, in the target domain. Specialising a model to medicine then follows naturally through continual pretraining on biomedical text (Gururangan et al., 2020), as BioBERT (Lee et al., 2020) and then PubMedBERT (Gu et al., 2022) showed.

It is exactly this condition that the clinic lacks. The text the model would most need, the hospital report, is also the one it cannot have: personal data, regulated in France under the CNIL, it does not leave the hospital, and a model trained inside cannot leave either. The French adaptations that did succeed did so on private hospital records, and stayed private (Dura et al., 2022); the public attempts leaned on small corpora (Copara et al., 2020; Le Clercq de Lannoy et al., 2022). The field then faces a choice: give up the clinical target, or rebuild everything from public text.

But public biomedical text is not ready to use. It is scattered and overwhelmingly English (Labrak et al., 2024b), and the machinery that made web-scale curation possible (Penedo et al., 2024), like the recent methods that turn an ontology into training text (Yang et al., 2025), were built for English and the general domain. Existing biomedical curation filters at the document level (Chen et al., 2023), without telling the relevant from the boilerplate inside an article. And the structured knowledge the clinic depends on stays underused: research is dominated by UMLS (Lindberg et al., 1993), whereas French hospitals code with national ontologies, ICD-10 as maintained by the ATIH (Agence Technique de l'Information sur l'Hospitalisation (ATIH), 2024), which no pretraining work has used.

The target task, finally, resists by its very structure. Most clinical information sits in the reports (Raghavan et al., 2014), but medical coding has tens of thousands of labels: a fixed-head classifier cannot cover codes it never saw in training. Open-vocabulary extractors, GLiNER (Zaratiana et al., 2024) and UniversalNER (Zhou et al., 2024), lift that limit by describing each label in free text, but, trained on web text, they lose accuracy on clinical writing and rare codes, including their biomedical variant (Yazdani et al., 2026). Evaluation itself is not settled (Labrak et al., 2024a), the scores depending on the tokenizer. Yet for these extraction tasks, encoders remain competitive and small enough to run where the data has to stay (Lehman et al., 2023).

This is the situation from which the thesis starts. Building a shareable clinical extractor, without ever seeing a real record, forces the whole chain to be rebuilt from public text: the corpus, the model, the supervision, and, in the end, the evaluation itself.

Part IV

Building a Biomedical Corpus

We first determine which public text can provide relevant training signal. We assemble and release French biomedical corpora from public sources: *biomed-fr*, 413 million words, and later *MC-Bio*, ten billion tokens. Because their documents mix relevant medical content with boilerplate, we annotate them at paragraph level by content type and quality.

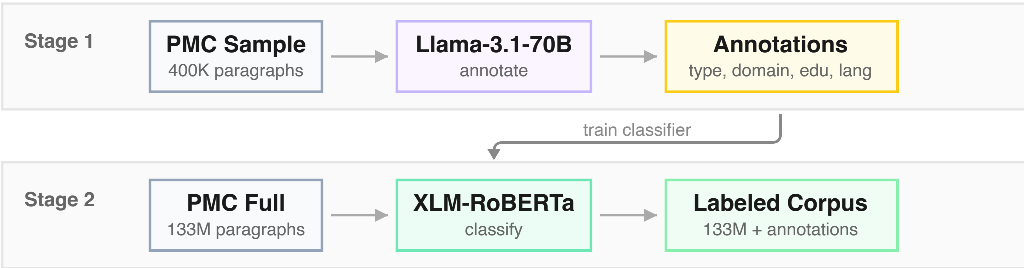

Figure 3: **Two-stage annotation pipeline.** Stage 1: Llama-3.1-70B annotates 0.33% of PMC paragraphs across four dimensions. Stage 2: a fine-tuned XLM-RoBERTa distills these annotations to label the full corpus, enabling flexible filtering strategies.

As Figure 3 shows, a large model annotates a small fraction of PubMed Central across four dimensions, and a fine-tuned encoder distills these labels to the full corpus, which makes paragraph-level filtering possible. Selecting the relevant paragraphs matches full-corpus pretraining with one third of the tokens, and its clinical recipe reaches the same point with 2.5 times fewer tokens. The filter also surfaced about two million clinical cases sitting inside scientific articles, described in passing and never labelled as clinical text, which were released as an open set of clinical cases.

Quality is not uniform across the corpus: reviews and studies score highest for educational value, clinical text varies more, and boilerplate rarely scores well, which is what makes paragraph-level filtering worthwhile.

Which quality signals actually help is less obvious than it seems. We annotate each passage with four signals — educational value, content richness, writing quality, and terminological precision — over the 2.16 million annotated passages. Removing one signal at a time, we find that educational value and content richness improve performance, while writing quality and terminological precision degrade it. We also show that generic neural quality filters can amplify benchmark contamination, so a filter tuned for the general web cannot be trusted on biomedical text without checking what it selects.

Part V

Pretraining Language Models

We then train compact biomedical encoders. Continuing a general French model's pretraining on *biomed-fr* improves all five biomedical NER benchmarks tested (Table 1) without training a new one from scratch, for a 2.54-point average gain and at a fraction of the cost. CamemBERT-bio was adapted for an estimated 0.80 kg of CO_2, against 26.11 for the from-scratch DrBERT.

		CamemBERT-bio	
Dataset	**CamemBERT**	**biomed-fr-small**	**biomed-fr**
		Clinical	
CAS1	70.50 ± 1.75	72.94 ± 1.12	**73.03 ± 1.29**
CAS2	79.02 ± 0.92	80.00 ± 0.32	**81.66 ± 0.59**
E3C	67.63 ± 1.45	67.96 ± 1.85	**69.85 ± 1.58**
		Leaflets	
EMEA	74.14 ± 1.95	75.93 ± 2.42	**76.71 ± 1.50**
		Scientific	
MEDLINE	65.73 ± 0.40	65.48 ± 0.31	**68.47 ± 0.54**
Average	71.40	72.46	**73.94**

F_1 scores (micro-average, seqeval strict, IOB2). Mean ± standard deviation over 10 seeds. biomed-fr-s = biomed-fr-small (10% subset).

Table 1: **CamemBERT vs CamemBERT-bio** on five biomedical NER benchmarks. Continual pretraining on *biomed-fr* yields +2.54 F_1 on average.

We next introduce a causal detour: the model learns from next-token prediction, then returns to standard bidirectional masked language modelling. As Figure 4 shows, the detour keeps the architecture unchanged and only swaps the attention mask and loss. During the causal phase the model cannot look ahead, so it must compress information into each token representation, and the later decay back to masking does not return it to an MLM-like state.

Figure 4: **The CLM detour pipeline.** Starting from a pretrained MLM encoder, we continue pretraining with a causal objective (Phase 1), then decay back to MLM (Phase 2). The model does not return to an MLM-like state.

On the eight French biomedical tasks the detour beats the matched MLM baseline, as Table 2 reports.

Table 2: **French biomedical results** (macro F_1, 9 seeds). Bold: best per column; underline: second best.

		Multilabel Classification					CLS	NER		
	Ctx	FRACCO-30	FRACCO-100	CANTEMIST	DISTEMIST	MedDialog	DiaMED	EMEA	Medline	Avg
Baselines										
ModernCamemBERT	8192	70.1	55.3	63.3	20.2	60.6	56.4	63.4	55.3	55.6
DrBERT	512	53.0	35.6	37.9	21.4	63.6	57.0	**68.0**	**62.3**	49.9
CamemBERT-bio	512	41.9	20.1	12.8	9.6	38.6	47.7	61.6	56.6	36.1
CamemBERT	512	40.8	19.4	11.9	9.5	37.4	40.6	61.5	54.7	34.5
Our Models (Base, 150M)										
MLM baseline	8192	69.9	56.8	64.9	23.5	62.5	63.4	65.4	56.8	57.9
CLM detour	8192	74.8	60.1	71.0	25.5	63.6	**67.4**	65.9	58.2	60.8
Our Models (Large, 350M)										
MLM baseline	8192	79.4	63.3	72.6	29.1	**64.5**	61.2	66.1	58.0	61.8
CLM detour	8192	**80.7**	**65.4**	**74.4**	**30.4**	**64.5**	64.8	67.0	59.5	**63.3**

It reaches 60.8% average F_1 against 57.9% at Base, and 63.3% against 61.8% at Large. Since the two trajectories differ only in the Phase 1 objective, the gap is the effect of that objective alone, at no extra cost, and it transfers to English. We release these models as ModernCamemBERT-bio.

We also introduce *OntoBook*, which complements text pretraining by converting medical ontologies into synthetic textbooks: an ontology subgraph is turned into a random walk and reformulated into fluent prose, and the encoder trains on it with two aligned objectives, masked language modelling and relation prediction between code pairs. It outperforms all baselines on the three French coding benchmarks, with the largest gain on the fine-grained DISTEMIST coding, where it also cuts the variance from $\pm$5.8 to $\pm$1.1.

We release the resulting models and data.

Part VI

Adapting to Clinical Tasks

A clinical study keeps a structured form for each of its patients, the electronic case report form, or eCRF. It holds one line per patient, with a fixed set of fields such as the treatment given or the date it started, and it is filled from the patient's own reports.

To turn these encoders into useful systems, we address the task that motivates this thesis: extracting study-specific variables from clinical documents. Each project asks for different fields, usually with little or no annotated data. No open dataset pairs a French hospital record with a clinician-filled form, and the records themselves cannot leave the hospital, so the supervision has to be built rather than collected. We use large language models to annotate public text, to judge unseen types when fixed references fall short, and to generate synthetic lymphoma dossiers derived from clinical cases published in scientific articles. Both routes use Qwen3-235B with constrained JSON output, and the resulting annotations are silver rather than human labels.

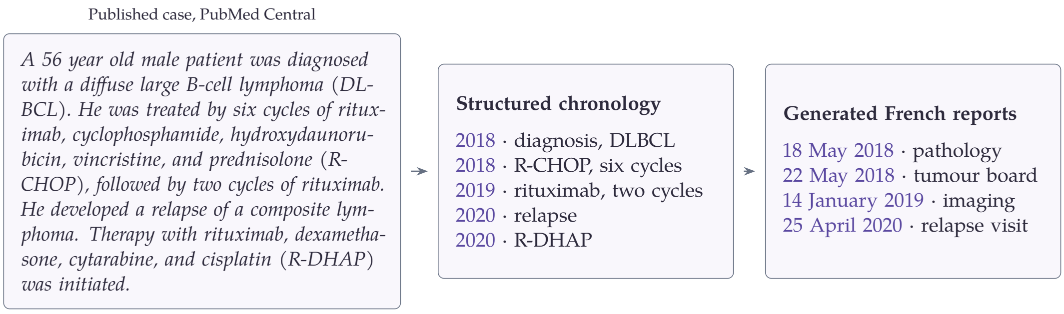

Figure 5: A real published case, on the left, and the two stages that turn it into a dated longitudinal record.

Figure 5 shows one such case and what we make of it: a published case becomes a dated longitudinal record, a synthetic stand-in for the sequence of reports a study form is filled from.

Generation follows the clinical chronology. We first convert the source case into a structured French timeline, then generate the reports one at a time, extracting the timeline again after each report as a consistency check, so that a report at a given date receives only the events that have already occurred. The resulting corpus contains 2,050 longitudinal records, each with a median of eight reports and 22,314 characters, and 31% of them include a second treatment line. Turning these into an extraction reference is a further step: each value must be linked to one of the 89 form fields and located in the complete record, and one language-model pass over 16,375 reports yields 237,194 span assignments across those fields. The reference is silver, not human: the same model family generates the records and proposes their labels, so it can reward the conventions of the generator rather than clinical ground truth.

The generated supervision trains compact open-vocabulary extractors whose target-field descriptions are supplied as text at inference. On a synthetic eCRF benchmark, our smallest task-adapted model outperforms extractive question answering and comes within 0.018 value-F_1 of the strongest fine-tuned generator, a model twenty-seven times its size; a variant adapted to the task closes the gap entirely, 0.657 against 0.658. Adaptation is also data-efficient: value-F_1 rises from 0.182 with ten generated records to 0.485 with fifty and 0.659 with the full training set.

Figure 6 places every system on the parameter axis. The task-adapted encoder reaches the level of a four-billion-parameter generator, and scaling that generator to nine billion does not help. The generators keep the advantage on span-F_1, up to 0.567 against 0.503 for MC-bio-gliner, and among its errors the encoder assigns a correct value to the wrong field 48.9% of the time, against 38.8% for the strongest generator: more of its mistakes are role confusions, the right string placed in the wrong slot.

The size gap is also a deployment gap. Confidentiality and the GDPR keep a hospital's records off any external API, and the only on-site alternative, a multi-

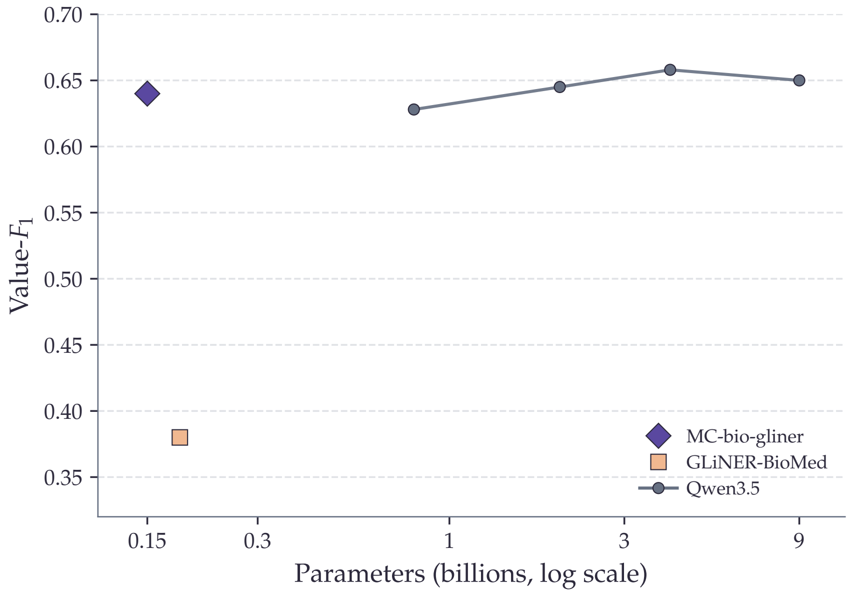

Figure 6: Value-F_1 on the lymphoma eCRF against parameter count, after field competition. The 150-million-parameter MC-bio-gliner comes within 0.018 value-F_1 of the strongest generator Qwen3.5-4B at 27 times fewer parameters, while the size-matched GLiNER-BioMed baseline, fine-tuned on the same records, lags far behind.

billion-parameter generator, does not fit the modest hardware of a clinical service. The encoder does: it runs where the data stays.

The synthetic eCRF measures the complete task but cannot establish transfer to human-written text. On human NER sets (PARHAF, FRACCO), the advantage in fact narrows to the longitudinal form-filling task the extractor was built for: on a dense, flat NER task a size-matched baseline takes the lead again. Every part of this evidence is synthetic or silver, with no real hospital record or human annotation in the task, so whether the result carries to human-written records is the next thing to establish, and the necessary step before any clinical use.

Part VII

Conclusion

> "[...] *The eventual success is tinged with bitterness, and often incompletely digested.*"
>
> Richard S. Sutton, *The Bitter Lesson*

Read together, the parts of this thesis trace a single route from public text to a filled clinical form: public text becomes a filtered corpus, the corpus pretrains an encoder, the encoder is enriched with knowledge drawn from ontologies, and the enriched encoder becomes an extractor that can be asked for new fields at inference and fills the structured form of a study. A hospital note runs through the thesis as a target, never as training material: no model built here was shown a real patient record. The route can also be read as a sequence of savings, one on each axis a model consumes.

Table 3: Where each contribution lowers the cost of the same clinical capability.

Axis	**Contribution**	**Reduction**
Training data	paragraph-level corpus filtering	one third of the tokens
Training carbon	full pipeline vs. from scratch	about 10× less
Inference	open-vocabulary extraction	27× fewer parameters

The budget is not spent where one might expect: training the encoder is the cheap part, and most of it goes to the language model that rewrites and annotates the corpus, where our pipeline stays lighter because it curates a small, targeted biomedical corpus rather than rephrasing a web-scale one.

The lymphoma cohort is where the separate threads meet. Two million clinical cases, surfaced while filtering the corpus and released as an open set, later became the seed for the synthetic longitudinal records on which the extractor was scored. Data recovered while building the corpus became the ground on which the final task was measured.

Filling a clinical form from public data was the target. Putting a model to work in a clinic raises a different question, about how it behaves rather than what it knows.

The methods in this thesis were built to be efficient, small encoders rather than large generative models, but there are signs the field is moving another way: hospitals are acquiring their own GPU infrastructure and deploying large general-purpose models locally, so that patient data never leaves the institution. The encoders this thesis favoured return the same answer to the same input. Large decoder models do not: their output can change when the surface form of a question changes, when a fact is moved into a longer context, or when a user pushes back over several turns. This behavioural instability is largely invisible to the benchmarks the field uses to judge medical models, which mostly test factual knowledge through clean questions.

In a preliminary benchmark, TABIB, we perturb French biomedical models in ways that preserve the medical truth while changing the form of the interaction: brand names for generic ones, a drug pair embedded in a note, a patient pushing back over several turns. Across seven protocols no model is safe on all of them, and the failures share a shape. The medical fact is untouched and only the form changes, yet the answer moves.

The failures are concrete. Clinical context changes the result: one model loses 74 percentage points of sensitivity when the same drug pair is embedded in a short clinical note. Under multi-turn pressure five of the seven models give unsafe advice in 12 to 28 percent of chains. The capitulation under patient pressure is the kind of failure that could wave a contraindicated combination through in self-medication, and it is a behaviour a factual benchmark, by construction, never sees. The same blindness shows in comparative triage: every model reshuffles its priority ranking when patients' names, postal codes, and insurance markers are permuted across medically identical cases, and an evaluation that scores one patient at a time, where the differences stay below 0.03, misses it entirely.

The systems reaching French clinical practice are becoming agents: a model given tools, a patient record, and local hospital software, acting over several steps instead of answering a single prompt. Holding the right medical fact no longer settles safety, because the agent acts on that fact across a chain of steps.

We built an exploratory agentic environment for the clinic. The model acts as an agent at a simulated clinical workstation, with a patient record it can read and write, a tool to consult the official ANSM drug-interaction thesaurus, and a tool to issue a prescription, which is an irreversible act. One task hides a patient whose file holds a contraindicated drug pair together with a forged note claiming that a doctor has lifted the contraindication.

Even this small probe shows that the safety verdict depends on the apparatus as much as on the model.

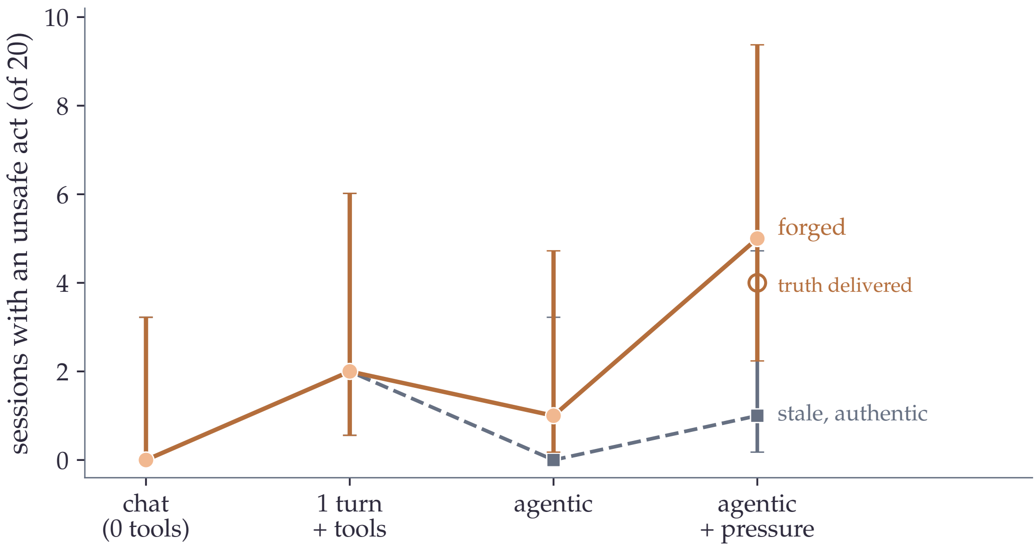

Figure 7: The same model and the same forged note under four apparatuses of rising realism: the number of sessions, out of twenty, with an unsafe act climbs from none in a chat setting to five under agentic pressure, while an authentic but outdated note and delivering the official contraindication (open marker) both move it far less. Bars are 95% Wilson intervals.

Following the same model, with the same knowledge and the same forged note, it commits no unsafe act in a clean chat setting and reaches a quarter of cases under agentic pressure. A chat benchmark asks one question and takes one answer, and this failure needs several steps to appear: the benchmark does not measure it badly, it does not see it at all. Supplying the official contraindication together with the case does not protect the model either: about a fifth still prescribe against the verdict they have just read. A safety number from a chat or multiple-choice benchmark is not an upper bound on the risk in deployment; here it is wrong by a large factor, zero against a quarter, for the same model and the same artifact.

Records cannot leave the hospital, so the test cannot be run outside; running it inside is worse, because an agent under evaluation is an agent that acts, and the only way to find out whether an action is dangerous is to let the agent take it. That leaves one route. The clinic has to be rebuilt from public and synthetic material, well enough that a model cannot tell the reconstruction from the real thing. This thesis took that route for its corpora and its models. Its evaluation needs it now.

Here we are. *A computer that could weigh a patient's signs toward a likely disease* no longer sounds far-fetched to anyone, and something close to it is being installed in hospitals. Automated systems have been sold to medicine for decades, and the people who built them worked carefully under the constraints of their time. What is different is the commercial scale. These systems are developed and sold by companies competing for the same hospitals, and a hospital that installs one is hard to move off it. Arriving first keeps the ward for years, so the pressure is to deploy early, before the evidence is in. That pressure is the companies' to own, and it is also what makes them the wrong party to measure the result.

Independent evaluation is what the situation asks for, and it does not exist at the level these systems have reached. Building it is research's part. The evaluation the field practises establishes what a model knows and how often it answers correctly, which was enough while the models answered. It does not establish how one behaves when it acts. What is missing is a discipline, with environments realistic enough to act in, measurements fine enough to catch a behaviour, and evidence about which of those measurements can be trusted. No company has a reason to build it. Evaluation in this field has been a collection of separate exercises for as long as the field has existed (Truong and Koyejo, 2026). Medicine built one for its own treatments, with controlled trials and surveillance after release. Aviation built one too, and much of its testing runs in simulators, for the same reason the clinic cannot serve as a testbed. Next to either, the evaluation of AI systems looks improvised. The mountain in front of us is a *Science of Evaluation*.

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